

Khalil Lagha

M2 student — Intelligent Systems for Automotive & Aerospace (SIAA) · Université Paris-Saclay

Ranked 2nd of the M1 EEA cohort (15.43/20). Control (PID, LQR, Kalman, FOC), Matlab/Simulink validation, power electronics for electric vehicles — from the lab (GIPSA-lab) to the industrial field. Work-study rotation: 2 days company / 3 days university.

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- Authorized to work in France



01 Experience

- Research Intern · GIPSA-lab, Grenoble** 04 – 07/2026 · 3.5 months
 - Modeled a PEMFC fuel cell in Matlab/Simulink — achieved **6%** dynamic and **1%** static error.
 - Tuned a **DC-DC boost converter** PI regulation loop to optimize efficiency; validated first in simulation, then on a real test bench.
- Intern · SLB (Schlumberger)** 07/2025 · 1 month
 - Authored technical syntheses in English in an international industrial environment.
- Power Electronics & Automation Intern · EURL Lagha** 01/2024 · 15 d + 09/2024 · 1 month
 - Contributed to the electronics of real industrial equipment: a **buck DC-DC converter** (50 V → 0–48 V, **25 A**) and 4 transformer-rectifiers (**100 V DC / 80 A**, ripple < 5%).
 - Co-designed the supervision HMI (**Siemens TIA Portal**, Ladder PLC); configured industrial control applications.
 - Produced the electrical schematics in **EPLAN** — direct interface with manufacturing.
- Power Grid Intern · Sonelgaz** 03/2024 · 15 d · observation
 - Analyzed substation structure and the grid voltage chain (**400/220 kV → 400/230 V**); surveyed an MV/LV substation on site.

02 Projects

Field-oriented control (FOC) — induction motor · Matlab/Simulink
Park transforms, PID controllers, observers, **Kalman filter**; cascaded current/speed loops — the core of electric-vehicle traction.

Neural network from scratch (MLP) · Matlab · open source (MIT)
Multilayer perceptron and backpropagation hand-coded, no toolbox — debugged, tested and published: github.com/KhalilEllahLAGHA/mlp-backpropagation-matlab.

Adjustable PWM generator · 555 timer + D flip-flops · Proteus
Adjustable frequency and duty cycle — simulated in Proteus then **built and fine-tuned on real hardware**: a complete electronics-diagnostics workflow.

Cascade control of a DC motor · Simulink · three-phase thyristor bridge
Outer speed loop, inner current loop; voltage control.



Work-study via CFA EVE
Université Paris-Saclay

2nd
of the M1 EEA
cohort (UGA)

15.43
M1 annual average
/20

C2
English

03 Education

- M2 SIAA — Intelligent Systems for Automotive & Aerospace** 2026 –
Université Paris-Saclay (Évry) · work-study
- M1 EEA (EECS)** 2025 – 26
Université Grenoble Alpes · 2nd in class
- Engineering cycle, Electrical Engineering** 2021 – 25
École Nationale Polytechnique d'Alger · 4 of 5
years completed; prep classes: top-ranked
- Baccalauréat in Mathematics** 2021
Highest honours — 16.93/20

04 Skills

Control: PID, LQR, Kalman

Embedded: C/C++, Arduino

Motor control (FOC)

EV power electronics

PLC / HMI / SCADA

AI: NN, GA, data fusion

Python (OOP)

Electrical CAD: EPLAN, AutoCAD

05 Software

Matlab/Simulink · Python · C/C++ · Arduino · Proteus ·
LTspice · Siemens TIA Portal · Step7 · EPLAN · AutoCAD
Electrical · SolidWorks · Git/GitHub

06 Certifications

SLB NEST & SIPP Level 2 — 07/2025 (HSE)

07 Languages

French

C1

English

C2

08 Soft skills

Fast learner

Rigorous

Organized

Problem solving

Comfortable with tools

Portfolio : khalilellahlagha.github.io/siaa/ · Code :
github.com/KhalilEllahLAGHA · IEEE Student Branch ENP · VIC